

Rings and Subrings

1. Rings: *Give an example for each of the following.*
 - a. A ring with identity
 - b. A ring without identity
 - c. A commutative ring
 - d. A non-commutative ring
 - e. A subring of $\mathbb{Z}$, $\mathbb{Z}_6$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$.
2. Prove that $5\mathbb{Z}$ is a ring.
3. Prove that the set of nonnegative integers is not a subring of $\mathbb{Z}$.
4. Prove that $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$ is a subring of $\mathbb{C}$.
5. Let R be a ring with subrings A and B . Prove that $A \cap B$ is also a subring.

Integral Domains and Fields

6. Integral domains and Fields: *Give an example for each of the following.*
 - a. An integral domain.
 - b. A ring which is not an integral domain.
 - c. A field.
 - d. A ring which is not a field.
 - e. An integral domain which is not a field.
 - Sets $\supseteq$ Groups $\supseteq$ Rings $\supseteq$ Integral Domains $\supseteq$ Fields
7. Let R be an integral domain and suppose that $ax - bx = 0$. Prove that if $a \neq b$ then $x = 0$.
8. Let $R = \mathbb{Z}_{10}$
 - f. Prove that R is not an integral domain.
 - g. Prove that R is not a field.
9. Let $R = \mathbb{Z}_5$
 - a. Prove that R is a field.
 - b. Prove that R is an integral domain.

Ideals

10. Ideals: *Give an example for each of the following.*
 - a. A ring and an ideal in that ring.
 - b. A ring and a subring which is not an ideal.
 - c. A ring other than $\mathbb{Z}$, and a principal ideal in that ring.
 - d. A ring, and a prime ideal in that ring.
 - e. A ring, and a maximal ideal in that ring.
11. Prove that for any ring R , 0_R is an ideal of R .
12. Prove that $\langle x - 1 \rangle$ is an ideal in $\mathbb{R}[x]$.
13. Find all the ideals and maximal ideals of $\mathbb{Z}_{12}$ (use a lattice).

Homomorphisms and Isomorphisms

14. Homomorphism and Isomorphisms: *Give an example for each of the following.*

- a. A 1-1 function between rings.
- b. A function between rings which is not 1-1.
- c. An onto function between rings.
- d. A function between rings which is not 1-1.
- e. A homomorphism between rings.
- f. A function between rings which is not a homomorphism.
- g. An isomorphism between rings (other than the identity function).
- h. A homomorphism between rings which is not an isomorphism.
- The first isomorphism theorem.

15. Prove that the rings $\mathbb{R}$ and $\mathbb{Z}$ are not isomorphic.

16. Prove that $\mathbb{Z}_m$ is not isomorphic to $\mathbb{Z}_n$ if and only if $n \neq m$.

17. Let $f: R \rightarrow S$ be a ring homomorphism.

- i. What is $f(0_R)$?
- j. Suppose that $f(a) - f(b) = 0$ find $f(a - b)$.
- k. Suppose that A is a subring of R . Is $f(A)$ a subring?

Factor Rings

18. Ideals: *Give an example for each of the following*

- a. A factor ring.
- b. Two elements in the factor ring above.
- c. An abelian factor ring.
- d. A factor ring with identity.

19. Let $D = \{x: x = 2n + 1 \text{ for } n \in \mathbb{Z}\}$, the set of odd integers.

- a. Prove that D is not an ideal in $\mathbb{Z}$.
- b. Prove that $\mathbb{Z}/D$ is not a ring.

20. Let $E = \{x: x = 2n \text{ for } n \in \mathbb{Z}\}$.

- a. Prove that E is an ideal in $\mathbb{Z}$.
- b. Prove that $\mathbb{Z}/E$ is a ring.

21. Let $R = \mathbb{Z}/\langle 20 \rangle$. Give two examples of distinct elements in R , and show that they are distinct.

22. Prove that $\mathbb{Z}_{10}/\langle 2 \rangle$ is a field.

23. Prove that $\mathbb{Z}[x]/\langle x \rangle$ is an integral domain but not a field.

24. Let $f: \mathbb{Z} \rightarrow \mathbb{Z}_8$. Find the kernel of f .

25. Define $f: \mathbb{Z} \rightarrow \mathbb{Z}_8$ as $f(x) = x \bmod 8$. Show that $\mathbb{Z}/\mathbb{Z}_8$ is isomorphic to $\mathbb{Z}_8$.

26. Recall that $\langle x \rangle$ is a principal ideal in $\mathbb{R}[x]$. The function $f: \mathbb{R}[x] \rightarrow \mathbb{R}$ defined by

$$f(a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0) = a_0$$

is an onto homomorphism. Prove that $\mathbb{R}[x]/\langle x \rangle$ is isomorphic to $\mathbb{R}$.